

PRIYANSHU RAI

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SUMMARY

Backend Software Engineer with strong hands-on experience building and operating production-grade distributed systems using Python, Django, Celery, Redis and Kafka. Specialized in multi-cloud database migrations, large-scale data validation tooling (DVT), high-volume ingestion pipelines, scraping systems and reliability engineering, with deep exposure to observability, CI/CD and on-call production debugging.

EXPERIENCE

Software Development Engineer — Park+

Mar 2025 – Present

- Designed and operated high-throughput backend services using Django, Celery, Redis and Kafka for challan verification, document generation, retry orchestration and webhook delivery across high-traffic production systems.
- Built a production-grade Database Validation Tool (DVT) to certify migration of 35M+ rows from GCP to Linode using secure SSH tunneling, time-windowed partitioning, row-count reconciliation, ID-range validation and SHA-256 hash-based row comparison with intelligent sampling.
- Engineered automated multi-table audit pipelines with rate limiting, fault-tolerant retries and structured exportable reports, ensuring strong data integrity guarantees and reducing manual verification effort by over 90%.
- Developed and maintained large-scale B2C and B2B ingestion systems with schema normalization, idempotent writes, failure classification and source-level observability for government and partner data integrations.
- Built resilient scraping and captcha workflows with session lifecycle management, proxy rotation and OCR fallback, significantly improving success rates in unstable external environments.
- Implemented production monitoring and reliability practices using Kibana, Grafana and New Relic, defining SLA dashboards, alerting pipelines and debugging playbooks for faster incident response.
- Deployed and debugged backend services using Docker, Jenkins and Kubernetes, resolving CI/CD failures, pod crashes and rollout issues across staging and production clusters.
- Optimized bulk CSV and batch processing pipelines handling 25L+ records by improving batching logic, transactional boundaries and retry strategies, increasing throughput and reducing production failures.

SKILLS

Languages: Python, Golang, C++, Java
Backend: Django, Django REST Framework, Redis, Celery, Kafka, REST APIs, Microservices
Databases: MySQL, MongoDB (Indexing, Query Optimization, Data Validation, Replication)
Infrastructure: Docker, Linux, Jenkins, Kubernetes, CI/CD, GCP, Linode
Observability: Kibana, Grafana, New Relic, Structured Logging, SLA Dashboards

PROJECTS

TradingBuddy — Real-Time Market Analytics Platform

- Built a TradingView-style market analytics platform supporting real-time price tracking, OHLC data, expiry-wise option data and historical analysis.
- Designed backend APIs for live and historical market data with optimized caching for low-latency reads.
- Implemented technical indicators such as EMA, RSI and MACD along with watchlists and alerting mechanisms.
- Architected the system to handle frequent updates and concurrent users while maintaining consistent performance.

TrackIt — Debt & Expense Management System

- Designed and developed a Django-based backend powering expense tracking, debt management and repayment planning with secure JWT-based APIs.
- Built analytics dashboards for monthly spends, dues, repayment history and exportable financial summaries.
- Implemented a priority-based debt repayment planner with automated monthly schedules.

EDUCATION

B.Tech — Computer Science & Engineering

CGPA: 7.41/10

MANIT, Bhopal — 2025

Jawahar Navodaya Vidyalaya (JNV), Sagar — Class XII: 93.6%